Inductive and Deductive Reasoning

Objective:

Use different types of reasoning to justify statements and arguments made about mathematics and mathematical concepts

DEFINITIONS

Logic – The science of correct reasoning.

Reasoning – The drawing of inferences or conclusions from known or assumed facts.

Reasoning can either be inductive or deductive.

Inductive and Deductive Reasoning

Inductive Reasoning

-process of reaching a general conclusion by examining specific examples.

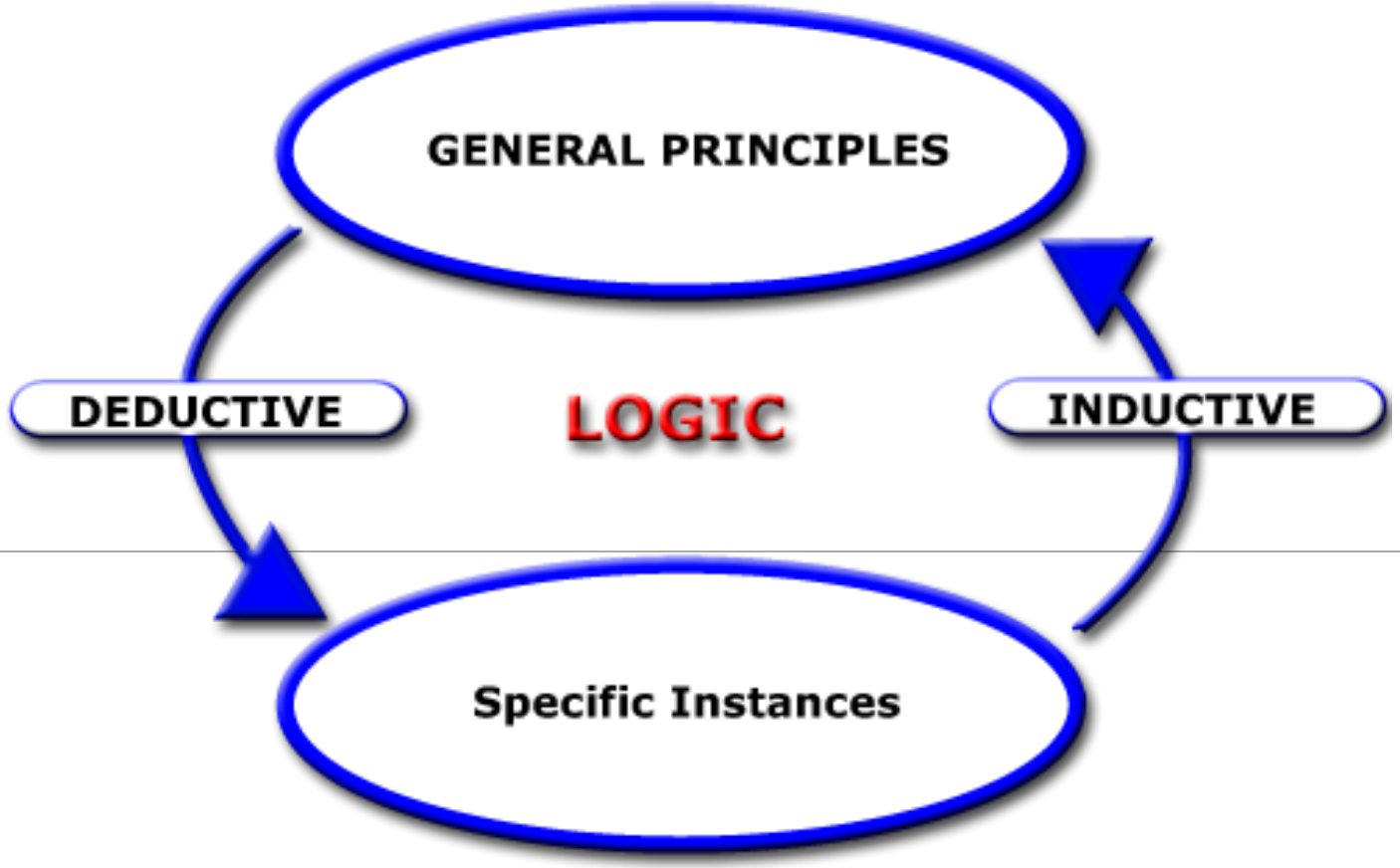
-uses patterns to arrive at a conclusion (conjecture)

Deductive Reasoning

-process of reaching a conclusion by applying general assumptions, procedures, or principles.

-uses facts, rules, definitions or properties to arrive at a conclusion.

Inductive and Deductive Reasoning



Inductive and Deductive Reasoning

In Deductive Reasoning,

TRUE PREMISE + **TRUE PREMISE** = **TRUE CONCLUSION**

while in Inductive Reasoning,

TRUE PREMISE + **TRUE PREMISE** = **PROBABLY TRUE CONCLUSION**

Inductive and Deductive Reasoning

In Deductive Reasoning,

TRUE PREMISE + **TRUE PREMISE** = **TRUE CONCLUSION**

All men are mortals.

Socrates is a man.

Socrates is a mortal.

Inductive and Deductive Reasoning

In Inductive Reasoning,

TRUE PREMISE + **TRUE PREMISE** = **PROBABLY TRUE CONCLUSION**

My professor wore blue blouse and blue shoes yesterday.

She wears a green blouse and green shoes today.

She will have the same color of blouse and shoes tomorrow.

Inductive or Deductive Reasoning?

Example:

My math teacher is skinny

My last math teacher was skinny

Therefore all math teachers are skinny

Answer: **Inductive**

Inductive or Deductive Reasoning?

Example:

My daddy has curly hair
My brother has curly hair

Therefore everyone I am related to has curly hair

Answer: **Inductive**

Inductive or Deductive Reasoning?

Example:

All oranges are fruits. All fruits grow on trees
Therefore, all oranges grow on trees.

Answer:

Deductive

Inductive or Deductive Reasoning?

Example:

Jimmy got bitten by a dog .

Jimmy is now scared of all dogs.

Answer:

Inductive

Inductive or Deductive Reasoning?

Example:

All even numbers are divisible by 2.

28 is even.

Therefore 28 is divisible by 2.

Answer:

Deductive

Inductive or Deductive Reasoning?

Example:

Two episodes of the Big Bang Theory made me laugh and laugh.

Therefore, The Big Bang Theory is always funny.

Answer: **Inductive**

Inductive or Deductive Reasoning?

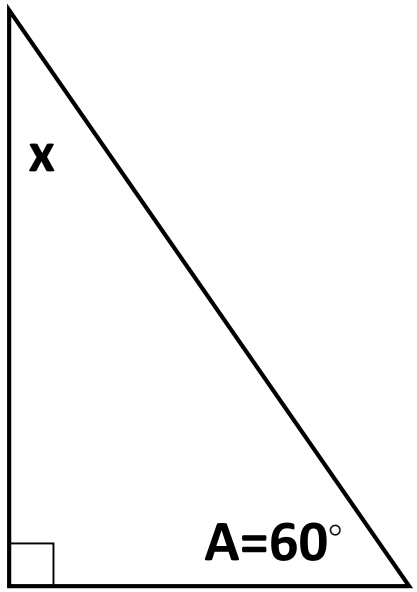
Example:

Jimmy named his kids Apricot-Mist,
Apricot-Mist Mu,
Apricot-Mist Mu Mu, and
Apricot-Mist Mu Mu Mu

Therefore Jimmy's next will be named
Apricot-Mist Mu Mu Mu Mu

Answer: **Inductive**

Inductive or Deductive Reasoning?



Example

The sum of the angles of any triangle is 180° and angle A in the figure is 60 degrees. Therefore angle x=180°-(90°+ A).

Answer: Deductive